Trade Study: IMU Sensor

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Decision: Adafruit 9-DOF Absolute (BNO085)

Problem: An IMU or inertial measurement unit, was chosen for this project's extra sensor. This is due to the need of knowing where and how fast the car is moving at a point in time. This allows us to match and calculate the speed and direction of a tracked object. The data received from an IMU is usually raw motion data, only giving acceleration or velocity at a point in time. Absolute orientation boards/sensors can calculate the current angle or position of the car in space, removing the need to write complex code to calculate these numbers on our own. Many IMUs also provide temperature data which can be useful in harsh environments where knowing the limits of the other sensors and components on the vehicle can guide mission decision making.

Assumptions:

- Uses GPIO pins on the Jetson.
- Uses a communication standard the Jetson supports.
- Is a 9-DOF sensor.
- Is able to be readily purchased. (Not used or limited stock)
- $20 is average price of a standard 9-DOF IMU
- Average data types are the standard raw Gyroscope, Accelerometer, Magnetometer outputs.

Rankings

Grades are ranked 0-5 for very below average, below average, average, above average, very above average.

		Adafruit 9-DOF Absolute (BNO055)	*Adafruit 9-DOF Absolute (BNO085)*	*Adafruit TDK InvenSense ICM-20948*
Criteria	**Weights**	**Grade**	**Grade**	**Grade**
Data Types (Amount)	0.35	4	5	3

Ease of Use / Compatibility	0.30	4	5	3
Power Management	0.20	5	5	4
Cost	0.15	2	2	3
Average Score	Max 5	3.75	4.25	3.25
Weighted Score	1.00	3.90	4.55	3.2

Breakdown

1st Choice: *Adafruit 9-DOF Absolute (BNO085)*

This IMU has the most amount of available sensor data that is calculated and available on board. This gives us many options for using the data to help track an object and track the positioning of the car. It scored 5 (Very Above Average) for this reason in Data Types. Due to the amount of pre-processed data types, absolute orientation data, and the STEMMA QT / Qwiic connectors making it easy to attach to the Jetson, a score of 5 was also given to Ease of Use / Compatibility. The BNO085 chip from CEVA Technologies includes many power modes that can be selected to disable and enable each internal sensor for reducing the power consumption or increasing the precision and polling rate of the sensors. This gives it a 5 in Power Management. Finally, the cost of $30 makes this a below average (~$15-$20) cost board, giving it a score of 3.

2nd Choice: Adafruit 9-DOF Absolute (BNO055)

This IMU shares many of the features with the BNO085, but lacks some of the extra pre-processed sensor data of the BNO085. This gives it a score of 4 in Data Types. For this same reason, usability slightly decreases to a 4, although all of the connectivity options remain the same. This chip also shares the many power management options as the 85, so the score remains at a 5. The cost of this board at $30 matches the 85, keeping the cost score at 2.

3rd Choice: Adafruit TDK InvenSense ICM-20948

This IMU only contains 3 (Gyroscope, Accelerometer, Magnetometer) raw sensors with no absolute or extra data. This gives it a Data Types score of 3, an average 9-DOF IMU. This harms usability, meaning we would need to create or use algorithms to determine absolute position data, and would lack temperature sensing. This gives it a Usability score of 3, also average for a typical 9-DOF IMU. The chip does contain some simple power management features, but it is already optimized for low power consumption. This gives it a score of 4 in Power Management. The cost at $20 makes it cheaper than

the other options, however it is about average for other simple IMUs, giving it a score of 3.